

Emacs Lisp

PENG

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1 Regular Expression

2 Automatic Indentation of code

Automatic indentation [1]:

- What is the right indentation of a line?
- When to reindent a line?

Indentation function:

- parsing forward from some safe starting point until the position of interest
- parsing backward from the position of interest

Generic indentation engine:

- CC-mode indentation code
- SMIE
- Tree-sitter

2.1 Simple Minded Indentation Engine

- operator precedence grammar
- unable to detect syntax error
- most programming languages cannot be parsed correctly by SMIE

Definition 2.1 (smie-setup).

(smie-setup grammar rules-function :forward-token :backward-token)

2.2 Tree-sitter

Parser-based indentation

3 Language Server Protocol

References

- [1] Bill Lewis, Richard M Stallman, and Dan LaLiberte. *GNU Emacs Lisp reference manual*. Free Software Foundation, 1998. ISBN: 1882114728.